

- (c) Find the voltage v_A with respect to the ground in the circuit.

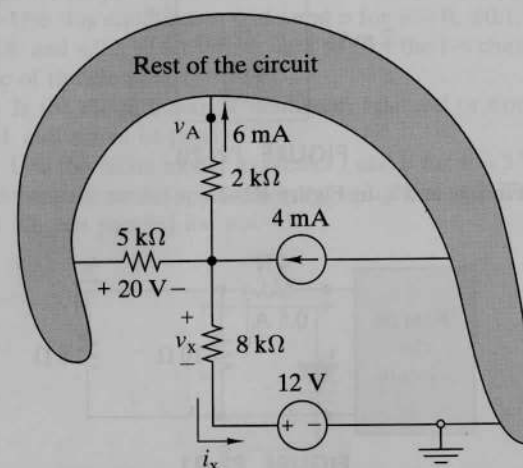


FIGURE P2-24

- 2-25 In Figure P2-25 $i_x = -0.5$ mA. Find the value of R .

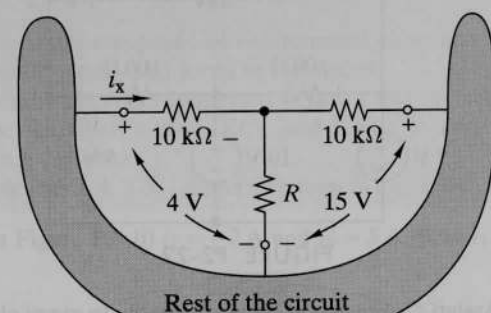


FIGURE P2-25

- 2-26 Figure P2-26 shows a resistor with one terminal connected to ground and the other connected to an arrow. The arrow symbol is used to indicate a connection to one terminal of a voltage source whose other terminal is connected to ground. The label next to the arrow indicates the source voltage at the ungrounded terminal. Find the voltage across, current through, and power dissipated in the resistor.

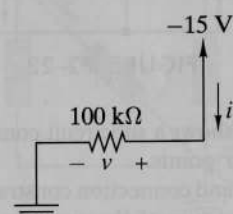


FIGURE P2-26

OBJECTIVE 2-4 EQUIVALENT CIRCUITS (SECT. 2-4)

- (a) Given a circuit consisting of linear resistors, find the equivalent resistance between a specified pair of terminals.
(b) Given a circuit consisting of a source-resistor combination, find an equivalent source-resistor circuit. See Examples 2-11, 2-12 and Exercises 2-7, 2-8, 2-9

- 2-27 Find the equivalent resistance R_{EQ} in Figure P2-27.

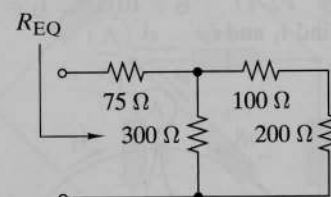


FIGURE P2-27

- 2-28 Find the equivalent resistance R_{EQ} in Figure P2-28.

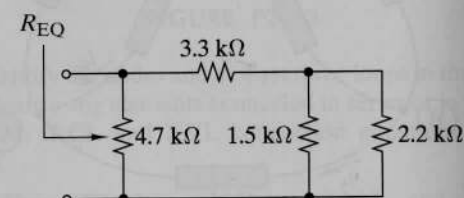


FIGURE P2-28

- 2-29 Find R_{EQ} in Figure P2-29 when the switch is open. Repeat when the switch is closed.

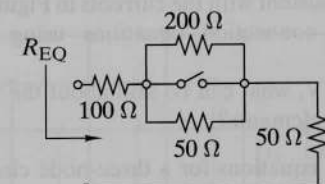


FIGURE P2-29

- 2-30 Show how the circuit in Figure P2-30 could be connected to achieve a resistance of 100 Ω, 200 Ω, 150 Ω, 50 Ω, 25 Ω, 33.3 Ω, and 133.3 Ω.

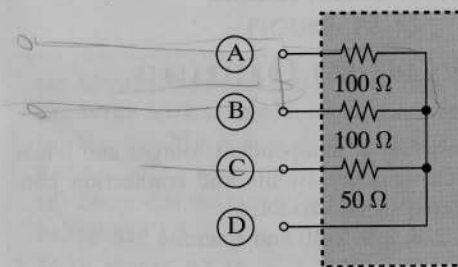


FIGURE P2-30

PROBLEMS

- 2-31 In Figure P2-31 find the equivalent resistance between terminals A-B, A-C, A-D, B-C, B-D, and C-D.

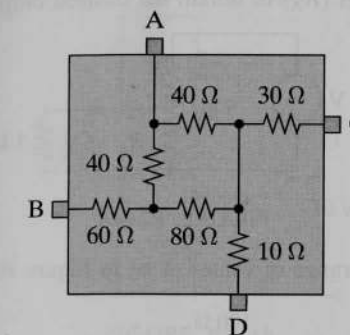


FIGURE P2-31

- 2-32 Select a value of R_L in Figure P2-32 so that $R_{EQ} = 25$ kΩ. Repeat for $R_{EQ} = 20$ kΩ.

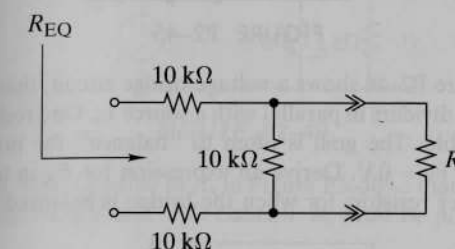


FIGURE P2-32

- 2-33 Using no more than four 1-kΩ resistors, show how the following equivalent resistors can be constructed: 2 kΩ, 500 Ω, 1.5 kΩ, 333 Ω, 250 Ω, and 400 Ω.

- 2-34 Find the equivalent practical voltage source at terminals A and B in Figure P2-34.

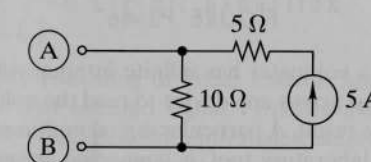


FIGURE P2-34

- 2-35 In Figure P2-35 the $i-v$ characteristic of network N is $v + 50i = 5$ V. Find the equivalent practical current source for the network.

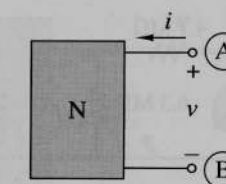


FIGURE P2-35

- 2-36 Select the value of R_x in Figure P2-36 so that $R_{EQ} = 60$ kΩ.

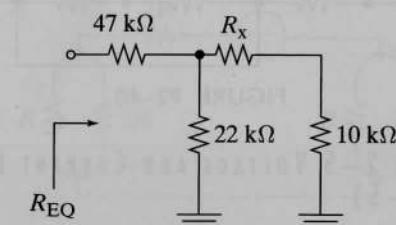


FIGURE P2-36

- 2-37 Two 10-kΩ potentiometers (a potentiometer is a variable resistor whose value between the two ends is 10 kΩ and between one end and the wiper—the third terminal—can range from 0 Ω to 10 kΩ) are connected as shown in Figure P2-37. What is the range of R_{EQ} ?

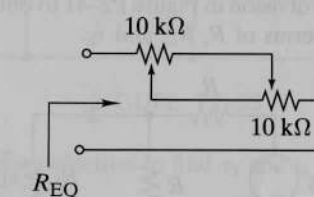


FIGURE P2-37

- 2-38 Select the value of R in Figure P2-38 so that $R_{AB} = R_L$.

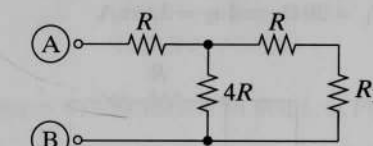


FIGURE P2-38

- 2-39 What is the range of R_{EQ} in Figure P2-39?

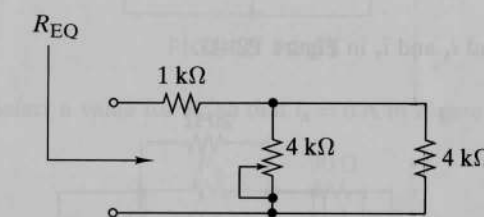


FIGURE P2-39

- 2-40 Find the equivalent resistance between terminals A and B in Figure P2-40.

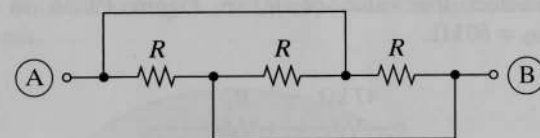


FIGURE P2-40

OBJECTIVE 2-5 VOLTAGE AND CURRENT DIVISION (SECT. 2-5)

- (a) Given a circuit with elements connected in series or parallel, use voltage or current division to find specified voltages or currents.
- (b) Design a voltage or current divider that delivers specified output signals within stated constraints.

See Examples 2-13, 2-14, 2-16, 2-17, 2-18 and Exercises 2-10, 2-11, 2-12

- 2-41 Use voltage division in Figure P2-41 to obtain an expression for v_L in terms of R , R_L , and v_S .

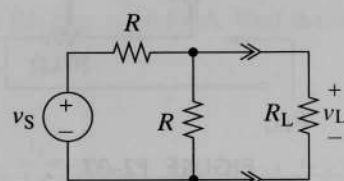


FIGURE P2-41

- 2-42 Use current division in Figure P2-42 to find v_L if $R = 10\ \Omega$, $R_L = 20\ \Omega$, and $i_S = 2.5\ \text{mA}$.

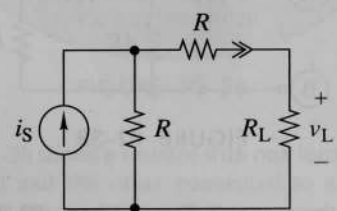


FIGURE P2-42

- 2-43 Find i_x and i_y in Figure P2-43.

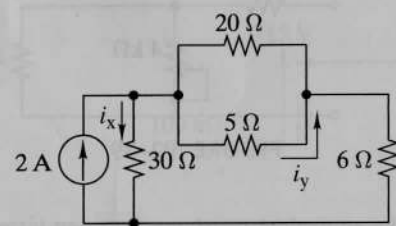


FIGURE P2-43

- 2-44 The $1\text{-k}\Omega$ load in Figure P2-44 needs 5 V across it to operate correctly. Where should the wiper on the potentiometer be set (R_X) to obtain the desired output voltage?

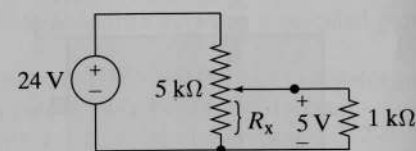


FIGURE P2-44

- 2-45 Find the range of values of v_O in Figure P2-45.

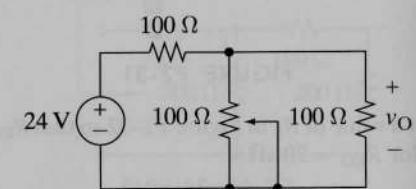


FIGURE P2-45

- 2-46 Figure P2-46 shows a voltage bridge circuit, that is, two voltage dividers in parallel with a source v_S . One resistor R_X is variable. The goal is often to "balance" the bridge by making $v_X = 0\ \text{V}$. Derive an expression for R_X in terms of the other resistors for when the bridge is balanced.

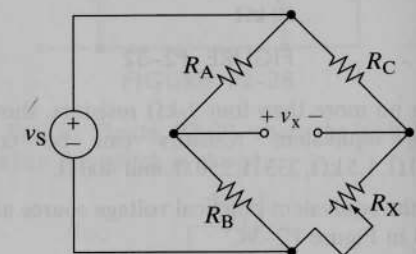


FIGURE P2-46

- 2-47 Ideally, a voltmeter has infinite internal resistance and can be placed across any device to read the voltage without affecting the result. A particular digital multimeter (DMM), a common laboratory tool, is connected across the circuit shown in Figure P2-47. The expected voltage was 5.73 V. However, the DMM reads 3.81 V. The large, but finite, internal resistance of the DMM was "loading" the circuit and producing a wrong measurement. Find the value of the internal resistance of this DMM.

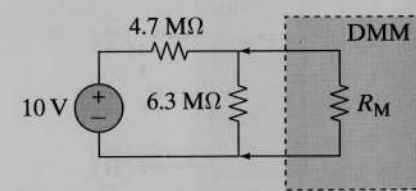


FIGURE P2-47

PROBLEMS

- 2-48 Select values for R_1 , R_2 , and R_3 in Figure P2-48 so the voltage divider produces the two output voltages shown.

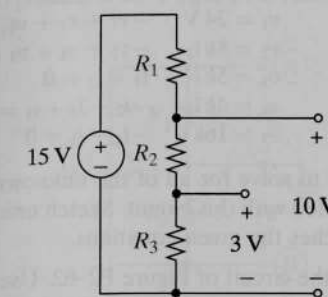


FIGURE P2-48

- 2-49 Select a value of R_X in Figure P2-49 so that $v_L = 3\ \text{V}$.

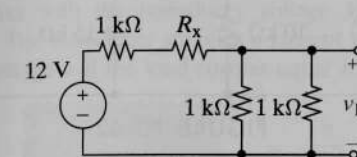


FIGURE P2-49

- 2-50 Select a value of R_X in Figure P2-50 so that $v_L = 2\ \text{V}$. Repeat for 4 V and 6 V. **Caution:** R_X must be positive.

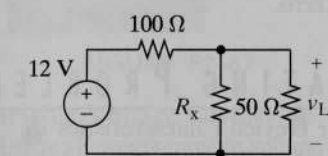


FIGURE P2-50

OBJECTIVE 2-6 CIRCUIT REDUCTION (SECT. 2-6)

Given a circuit consisting of linear resistors and an independent source, find selected signal variables using successive application of series/parallel equivalence, source transformations, and voltage/current division. See Examples 2-20, 2-21, 2-22, 2-23 and Exercises 2-13, 2-14, 2-15

- 2-51 Use circuit reduction to find v_x and i_x in Figure P2-51.

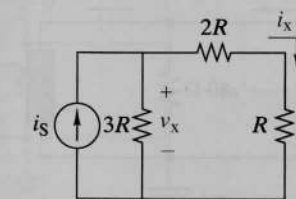


FIGURE P2-51

- 2-52 Use circuit reduction to find v_x and i_x in Figure P2-52.

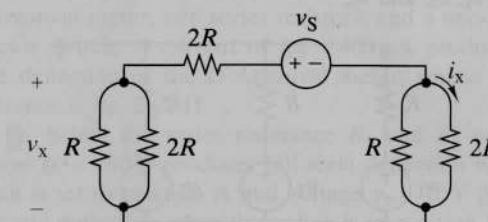


FIGURE P2-52

- 2-53 Use circuit reduction to find v_x and the power provided by the source in Figure P2-53.

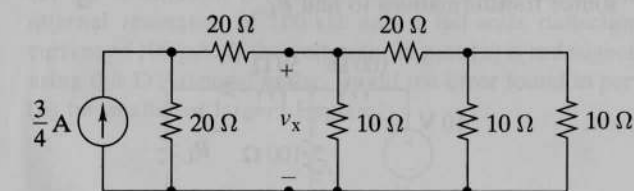


FIGURE P2-53

- 2-54 Use circuit reduction to find v_x and i_x in Figure P2-54.

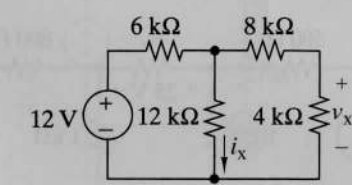


FIGURE P2-54

- 2-55 Use source transformation to find i_x in Figure P2-55.

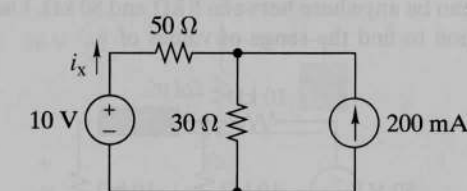


FIGURE P2-55

- 2-56 Select a value for R_X so that $i_x = 0\ \text{A}$ in Figure P2-56.

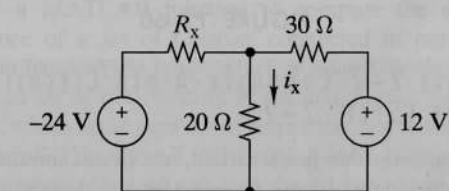


FIGURE P2-56

- 2-57 Use source transformations in Figure P2-57 to relate v_O to v_1 , v_2 , and v_3 .

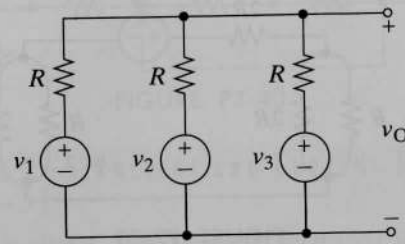


FIGURE P2-57

- 2-58 The current through R_L in Figure P2-58 is 40 mA. Use source transformations to find R_L .

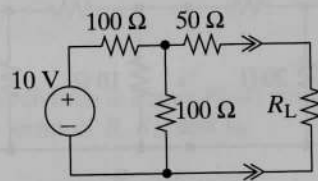


FIGURE P2-58

- 2-59 Select R_x so that 25 V are across it in Figure P2-59.

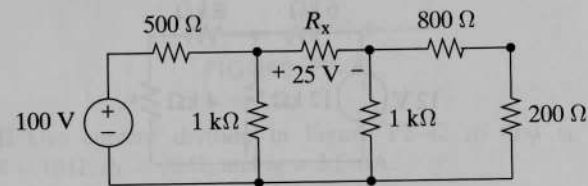


FIGURE P2-59

- 2-60 The box in the circuit in Figure P2-60 is a resistor whose value can be anywhere between 8 kΩ and 80 kΩ. Use circuit reduction to find the range of values of v_x .

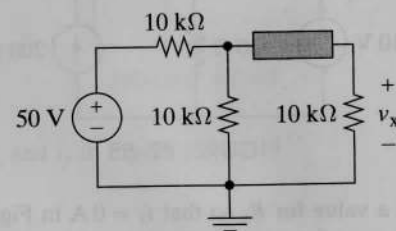


FIGURE P2-60

OBJECTIVE 2-7 COMPUTER-AIDED CIRCUIT ANALYSIS (SECT. 2-7)

Given an appropriate linear circuit, use circuit simulation and/or computational software to solve for the desired response. See Examples 2-24, 2-25 and Appendix B

- 2-61 A circuit is found to have the following element and connection equations:

$$\begin{aligned} v_1 &= 24 \text{ V} & -v_1 + v_2 + v_3 &= 0 \\ v_2 &= 8k i_2 & -v_3 + v_4 + v_5 &= 0 \\ v_3 &= 5k i_3 & i_1 + i_2 &= 0 \\ v_4 &= 4k i_4 & -i_2 + i_3 + i_4 &= 0 \\ v_5 &= 16k i_5 & -i_4 + i_5 &= 0 \end{aligned}$$

Use MATLAB to solve for all of the unknown voltages and currents associated with this circuit. Sketch one possible schematic that matches the given equations.

- 2-62 Consider the circuit of Figure P2-62. Use MATLAB to find all of the voltages and currents in the circuit.

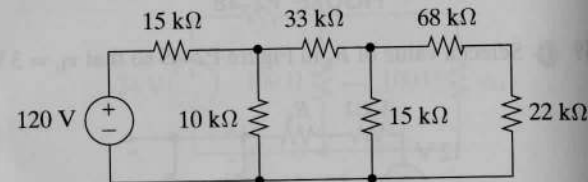


FIGURE P2-62

- 2-63 Consider the circuit of Figure P2-62 again. Use OrCAD to find all of the voltages and currents in the circuit.

- 2-64 Use OrCAD to find all the power dissipated or provided in the circuit of Figure P2-62. Verify that the total power in the circuit is zero.

INTEGRATING PROBLEMS

2-65 Nonlinear Device Characteristics A

The circuit in Figure P2-65 is a parallel combination of a 50-Ω linear resistor and a varistor whose i - v characteristic is $i_v = 2.6 \times 10^{-5} v^3$. For a small voltage, the varistor current is quite small compared to the resistor current. For large voltages, the varistor dominates because its current increases more rapidly with voltage.

- Plot the i - v characteristic of the parallel combination.
- State whether the parallel combination is linear or nonlinear, active or passive, and bilateral or nonbilateral.
- Find the range of voltages over which the resistor current is at least 10 times as large as the varistor current.
- Find the range of voltages over which the varistor current is at least 10 times as large as the resistor current.

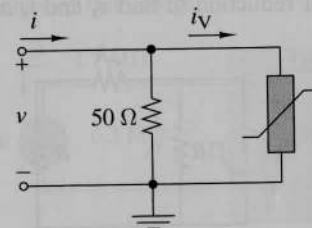


FIGURE P2-65

2-66 Center Tapped Voltage Divider A

Figure P2-66 shows a voltage divider with the center tap connected to ground. Derive equations relating v_A and v_B to v_S , R_1 , and R_2 .

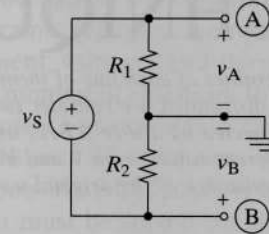


FIGURE P2-66

2-67 Active Transducer A

Figure P2-67 shows an active transducer whose resistance $R(V_T)$ varies with the transducer voltage V_T as $R(V_T) = 0.5V_T^2 + 1$. The transducer supplies a current to a 10-Ω load. At what voltage will the load current equal 100 mA?

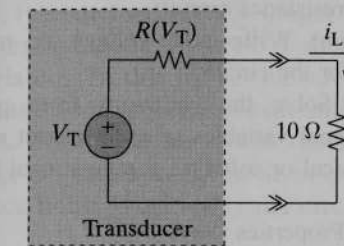


FIGURE P2-67

2-68 Programmable Voltage Divider A

Figure P2-68 shows a programmable voltage divider in which digital inputs b_0 and b_1 control complementary analog switches connecting a multitap voltage divider to the analog output v_O . The switch positions in the figure apply when digital inputs are low. When inputs go high the switch positions reverse. Find the analog output voltage for $(b_1, b_0) = (0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$ when $V_{REF} = 12 \text{ V}$.

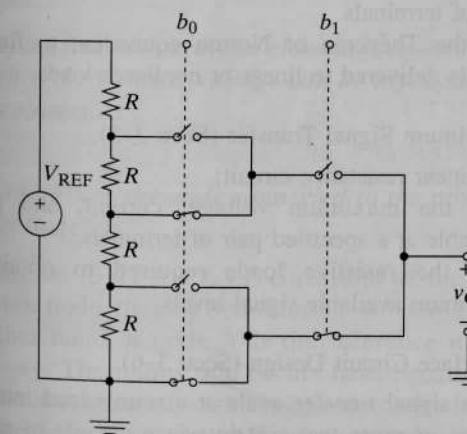


FIGURE P2-68

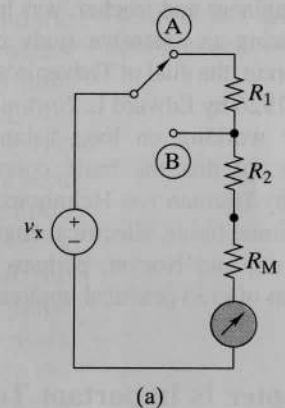
2-69 Analog Voltmeter Design (A, D, E)

Figure P2-69(a) shows a voltmeter circuit consisting of a D'Arsonval meter, two series resistors, and a two-position selector switch. A current of $I_{FS} = 400 \mu\text{A}$ produces full-scale deflection of the D'Arsonval meter, whose internal resistance is $R_M = 25 \Omega$.

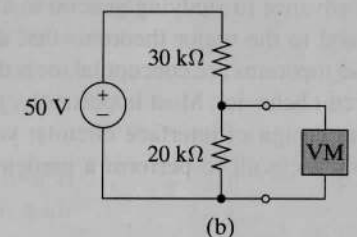
- (a) D Select the series resistance R_1 and R_2 so that a voltage $v_x = 100 \text{ V}$ produces full-scale deflection when the switch is set to position A and voltage $v_x = 10 \text{ V}$ produces full-scale deflection when the switch is in position B.

- (b) A What is the voltage across the 20-kΩ resistor in Figure P2-69(b)? What is the voltage when the voltmeter in part (a) is connected across the 20-kΩ resistor? What is the percentage error introduced in connecting the voltmeter?

- (c) E A different D'Arsonval meter is available with an internal resistance of 100-kΩ and a full-scale deflection current of 100 μA. If the voltmeter in part (a) is redesigned using this D'Arsonval meter, would the error found in part (b) be smaller or larger? Explain.



(a)



(b)

FIGURE P2-69

2-70 MATLAB Function for Parallel Equivalent Resistors A

Create a MATLAB function to compute the equivalent resistance of a set of resistors connected in parallel. The function has a single input, which is a vector containing the values of all of the resistors in parallel, and it has a single output, which is the equivalent resistance. Name the function "EQ parallel" and test it with at least three different resistor combinations. At least one test should have three or more resistor values.

- 3-30(a)** Use the superposition principle to find v_O in terms of v_1 , v_2 , and R in Figure P3-30.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

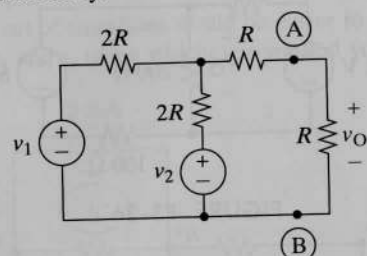


FIGURE P3-30

- 3-31(a)** Use the superposition principle to find v_O in terms of v_S , i_S , and R in Figure P3-31.
(b) Use MATLAB and node-voltage analysis to verify your answer symbolically.

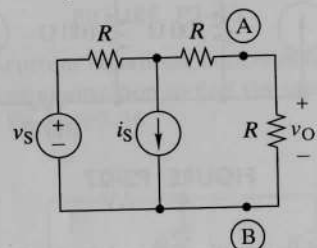


FIGURE P3-31

- 3-32** A linear circuit containing two sources drives a $100\text{-}\Omega$ load resistor. Source number 1 delivers 250 mW to the load when source number 2 is off. Source number 2 delivers 4 W to the load when source number 1 is off. Find the power delivered to the load when both sources are on. (Hint: The answer is not 4.25 W . Why?)

- 3-33** A linear circuit is driven by an independent voltage source $v_S = 10\text{ V}$ and an independent current source $i_S = 10\text{ mA}$. The output voltage is $v_O = 2\text{ V}$ when the voltage source is on and the current source off. The output is $v_O = 1\text{ V}$ when both sources are on. Find the output voltage when $v_S = 20\text{ V}$ and $i_S = -20\text{ mA}$.

- 3-34** A certain linear circuit has four input voltages and one output voltage v_O . The following table lists the output for different values of the four inputs. Find the input-output relationship for the circuit. Specifically, find an expression for v_O in terms of the four input voltages.

v_{S1} (V)	v_{S2} (V)	v_{S3} (V)	v_{S4} (V)	v_O (V)
2	4	-4	1	20
1	2	2	1.5	-4
1	4	2	2	-1
0	5	3	-1	3

OBJECTIVE 3-3 THÉVENIN AND NORTON EQUIVALENT CIRCUITS (SECT. 3-4)

Given a circuit containing linear resistors and independent sources,

- (a) Find the Thévenin or Norton equivalent at a specified pair of terminals.
 (b) Use the Thévenin or Norton equivalent to find the signals delivered to linear or nonlinear loads.

See Examples 3-13, 3-14, 3-15, 3-16 and Exercises 3-23, 3-24, 3-25, 3-26

- 3-35(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-35.

- (b)** Use the equivalent circuit found in part (a) to find i_L if $R_L = 47\text{ k}\Omega$.

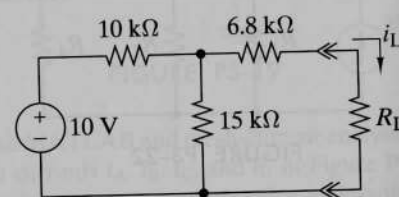


FIGURE P3-35

- 3-36(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.

- (b)** Use the equivalent circuit found in part (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .

- (c)** Check your answer to part (b) using current division.

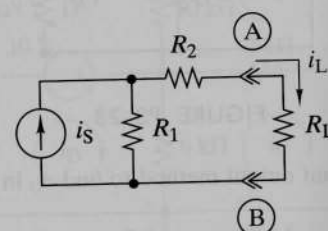


FIGURE P3-36

- 3-37** Find the Thévenin equivalent circuit seen by R_L in Figure P3-37. Find the voltage across the load when $R_L = 5\text{ }\Omega$, $10\text{ }\Omega$, and $50\text{ }\Omega$.

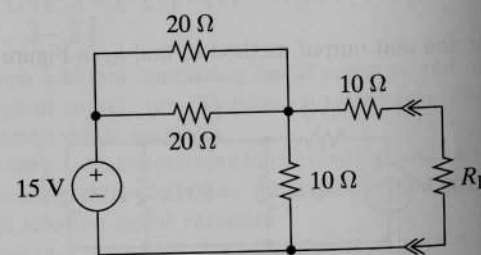


FIGURE P3-37

PROBLEMS

- 3-38** Find the Norton equivalent seen by R_L in Figure P3-38. Find the current through the load for $R_L = 4.7\text{ k}\Omega$, $15\text{ k}\Omega$, and $68\text{ k}\Omega$.

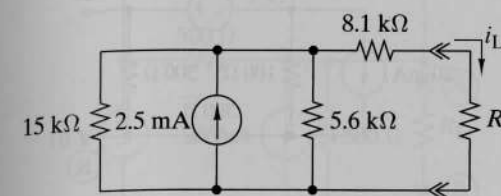


FIGURE P3-38

- 3-39** Find the Thévenin equivalent seen by R_L in Figure P3-39. Find the power delivered to the load when $R_L = 50\text{ k}\Omega$ and $200\text{ k}\Omega$.

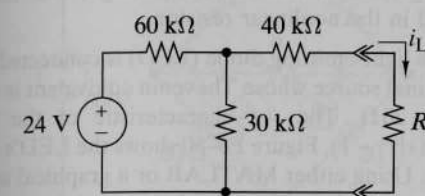


FIGURE P3-39

- 3-40(a)** Use OrCAD to find the Norton equivalent at terminals A and B in Figure P3-40. (Hint: Find the open-circuit voltage and short-circuit current at the requisite terminals.)

- (b)** Use the Norton equivalent circuit found in part (a) to determine the power dissipated in R_L when it is equal to $37\text{ k}\Omega$.

- (c)** Use OrCAD to simulate both the original and the Norton equivalent circuits with $R_L = 37\text{ k}\Omega$. Verify that the power dissipated by the load is the same in both situations.

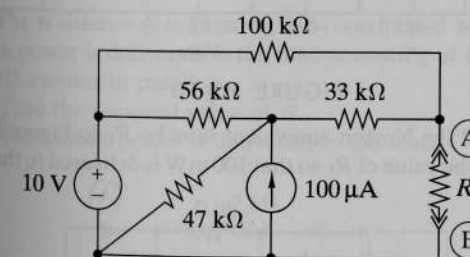


FIGURE P3-40

- 3-41** The purpose of this problem is to use Thévenin equivalent circuits to find the voltage v_x in Figure P3-41. Find the Thévenin equivalent circuit to the left of terminals A and B.

Find the Thévenin equivalent circuit to the right of terminals A and B. Connect these equivalent circuits together and find the voltage v_x .

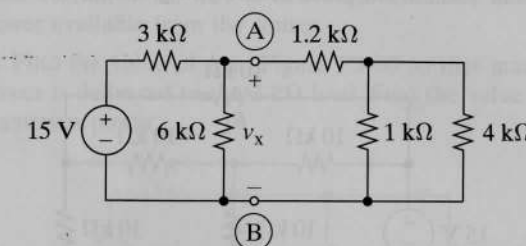


FIGURE P3-41

- 3-42** Assume that Figure P3-42 represents a model of the auxiliary output port of a car. The output current is $i = 1\text{ A}$ when $v = 0\text{ V}$. The output voltage is $v = 12\text{ V}$ when a $20\text{-}\Omega$ resistor is connected between the terminals. Suppose you wanted to charge a 12-V battery by connecting the battery at the port. How much current would the port deliver to the battery?

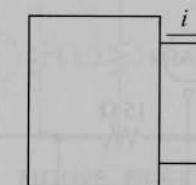


FIGURE P3-42

- 3-43** The i - v characteristic of the active circuit in Figure P3-42 is $5v + 500i = 100$. Find the output voltage when a $1\text{-k}\Omega$ resistive load is connected.

- 3-44** You have successfully completed Circuits I, and as part of an undergraduate work-study program your former professor has asked you to help her grade a Circuits I quiz. On the quiz, students were asked to find the power supplied by the source both to the $10\text{-k}\Omega$ load (R_L) and to the entire circuit as shown in Figure P3-44. Your professor asks you to help her by creating a grading sheet.

- (a)** Solve the quiz and establish reasonable A, B, C, D, and F cuts for incorrect solutions.

- (b)** A particular student correctly finds the Thévenin equivalent circuit seen by the resistive load and calculates the power to the load using V_T^2/R_L . He then does a source transformation, correctly finding the Norton equivalent of the circuit. He calculates the source power using $V_T \times I_N$. What grade would you give him?

- (c)** Another student finds $P_L = 5.625\text{ mW}$ and $P_S = -22.5\text{ mW}$, but provides no work to justify her answers. What grade would you give her?

- (d)** A third student first finds the Norton equivalent, and then finds the current through the load using a current